Practical 4: Crystallization

On successful completion of this practical, you should be able to:

- Use the TAS diagram to classify igneous rocks based on geochemical data
- Plot variation diagrams for suites of igneous rocks
- Use variation diagrams to interpret and predict crystallizing assemblages and how they change during fractionation

You will also practice the following skills:

- Describe the mineralogy and texture of an igneous thin section
- Classify igneous rocks by selecting an appropriate petrological classification scheme

Part 1 (60 minutes): Plotting and interpreting real geochemical data

Table 1 shows geochemical analyses of a suite of fine-grained igneous rocks collected from Thestareykir and Krafla in North Iceland, an area that is exploited for its geothermal energy.

Table 1: *Bulk rock compositions of igneous rocks from Theistareykir and Krafla, North Iceland, analysed by XRF spectroscopy. Major element oxides are report in weight %.*

	1	2	3	4	5	6	7	8	9	10
SiO_2	48.86	48.92	48.95	49.11	49.34	51.15	56.67	59.64	67.78	74.42
TiO_2	0.73	0.69	0.71	1.20	1.64	2.03	1.42	1.29	0.75	0.35
Al_2O_3	16.10	15.70	14.90	15.14	14.80	13.56	14.68	15.30	13.36	12.14
FeO	9.41	9.38	9.39	10.63	12.15	14.73	10.96	8.81	6.05	3.95
MnO	0.17	0.18	0.17	0.18	0.21	0.25	0.17	0.15	0.16	0.10
MgO	9.73	10.31	11.05	8.67	7.64	5.42	3.64	2.72	1.53	0.19
CaO	13.25	13.01	13.10	12.97	11.64	9.79	7.85	6.78	4.42	1.99
Na_2O	1.64	1.74	1.62	1.88	2.19	2.47	3.53	3.99	3.72	4.19
K_2O	0.06	0.05	0.05	0.11	0.22	0.40	0.76	0.96	2.06	2.60
P_2O_5	0.06	0.02	0.05	0.11	0.16	0.19	0.31	0.35	0.18	0.05

1) Classify magma compositions 6-10 using the total alkalis vs. silica (TAS) diagram on p.13 of the Practical Reference Handout.

   Rock 6 are basalts (and so are rocks 1-5). Rock 7 is a basaltic andesite. Rock 8 is an andesite. Rock 9 is a dacite. Rock 10 is a rhyolite.

2) Plot a variation diagram showing SiO_2 on the x axis vs. FeO on the y axis using the graph template provided. The first rock composition has been plotted for you.

   See plots in separate file.

3) Add an arrow to the x axis to show how SiO2 changes as fractional crystallization proceeds. Then comment on the trends you observe: How does the FeO content vary with increasing SiO_2?

   SiO_2 increases as fractional crystallization proceeds, so the arrow should point from left to right. The FeO content increases linearly until the SiO_2 content reaches just over 50 wt.%. Then there is a sharp change: the FeO content decreases as SiO_2 increases. The two segments of the trend are not quite linear.

4) These rocks are related to one another by fractional crystallization. What do the sudden changes in trend mean? Briefly inspect Table 1 and comment on how other major element oxide contents change as SiO_2 increases.

The change in trend must indicate a change in the crystallizing assemblage. In other words, each trend is associated with the crystallization of a different mineral (or group of minerals). Or, alternatively, the same minerals are crystallizing but in different proportions.

From the table: Fractional crystallization causes the Na_2O and K_2O contents to increase – this means that Na and K are not being taken up in the crystallizing minerals. The CaO content decreases during crystallization, so it must be being taken up in the crystallizing minerals (most probably into cpx and plagioclase). The TiO_2 content increases to ~2 wt.%, but then decreases – so a mineral that takes up Ti must start crystallizing when the SiO_2 content reaches ~55 wt.%. The Al_2O_3 content remains fairly constant during crystallization.

The next task is to determine what crystallising mineral(s) can produce this liquid line of descent. We will work on the graph of FeO vs. MgO that has been provided for you.

5) Before plotting anything, add an arrow to the x axis of the plot to show how MgO changes as fractional crystallization proceeds.

MgO decreases as crystallization proceeds.

Table 2 shows representative mineral compositions that are found in basalts from North Iceland.

6) Add an arrow to the x axis of the plot to show how MgO changes as fractional crystallization proceeds

MgO decreases as fractional crystallization proceeds, so the arrow should point from right to left. (This is different from the first plot, where crystallization proceeded from left to right.)

Table 2: *Representative compositions of olivine, plagioclase and clinopyroxene in basalts from Theistareykir and Krafla, North Iceland. Mineral compositions are reported in weight % oxide; a dash indicates that the oxide concentration is below the detection limit of the analytical technique.*

	Ol	Plag	Cpx
SiO_2	39.8	46.9	53.2
TiO_2	-	-	0.5
Al_2O_3	-	33.4	1.9
FeO	13.9	0.6	7.1
MnO	0.2	-	-
MgO	46.3	0.2	17.7
CaO	0.3	17.7	19.1
Na_2O	-	1.5	0.2
K_2O	-	-	-
P_2O_5	-	-	-

7) Add the compositions of olivine, plagioclase and clinopyroxene to the plot of FeO vs. MgO. Make sure each mineral is clearly labelled.

Add tie lines between each pair of minerals. You should end up with a triangle.

See plots in separate file.

8) We will now examine the liquid line of descent that links the magma compositions between ~10 and ~5 wt.% MgO. Add a linear line of best fit through these compositions, and use this best-fit line to predict the possible combinations of crystallizing minerals that could explain the chemical trend.

The back-projection of the trend that links these rock compositions intersects the plag-cpx and plag-ol tie lines, and crosses through the triangle. It doesn't intersect the cpx-ol tie line. So, the crystallizing assemblage could be any one of the following three options:

- olivine + plagioclase
- plagioclase + clinopyroxene
- olivine + plagioclase + clinopyroxene

There is not enough information to determine which of these three options is correct, but the most likely answer is the three-mineral assemblage.

9) Now look at the trend that links magma compositions between ~5 and 0 wt.% MgO. Can this trend be explained by crystallizing any combination of the three minerals on your plot?

No, it can't – because the trend through these magma compositions doesn't pass through the triangle or any of the tie lines linking olivine, plagioclase and cpx. This means that the crystallizing assemblage must change at ~5 wt.% MgO. This is consistent with the sudden change in trend.

10) What can you predict about the composition of the mineral that must join the crystallizing assemblage at ~5 wt.% MgO?

The trend through the low-MgO magma compositions shows a sharp decrease in FeO, so the new mineral must be Fe-rich. The most likely option is an oxide like magnetite (Fe_3O_4) or ilmenite ($FeTiO_3$). Table 1 shows that the TiO_2 content of the magmas starts to decrease at the same point the FeO contents start to decrease – so this suggests that ilmenite is the new mineral, because ilmenite crystallization will remove both Fe and Ti from the melt.

Table 3 shows the compositions of four phenocryst-rich picrites from North Iceland.

Table 3: *Bulk rock compositions of picrites from Theistareykir and Krafla, North Iceland, analysed by XRF spectroscopy. Major element oxides are report in weight %.*

	A	B	C	D
SiO_2	45.73	46.32	46.68	47.02
TiO_2	0.38	0.42	0.46	0.50
Al_2O_3	9.97	11.11	11.68	12.09
FeO	8.96	8.90	8.90	8.94
MnO	0.15	0.16	0.16	0.16
MgO	24.44	21.65	20.19	18.76
CaO	9.35	10.30	10.71	11.28
Na_2O	0.97	1.11	1.17	1.21
K_2O	0.02	0.01	0.01	0.01
P_2O_5	0.03	0.02	0.03	0.03

11) Add the four picrite compositions to your plot of FeO vs. MgO. Use a different symbol or colour.

See plots in separate file.

12) The most MgO-rich basaltic melt composition that has ever been measured in Iceland contains ~11 wt.% MgO. The picrites all contain >11 wt.% MgO. Fractional crystallization (removal of crystals) always causes MgO to decrease, so the picrite compositions cannot have been produced by fractional crystallization. What alternative process could produce the high-MgO picrites?

If it's not crystallization (removal of crystals) then it must be addition of crystals. Crystal accumulation and entrainment can happen if an erupting magma picks up crystals from the floor of the magma chamber before it erupts, or if the magma picks up crystals from old magma chambers as it makes its way to the surface. To explain the picrite compositions, the added crystals must be of MgO-rich minerals.

13) In light of your answer to (9), predict the abundance and types of phenocrysts you might expect to find in the picrite rocks.

The trend line through the picrites passes through the ol-cpx and plag-ol tie lines of the triangle, and through the centre of the triangle. So the magma must "pick up" a crystal assemblage that contains either:

- olivine + plagioclase
- olivine + clinopyroxene
- olivine + plagioclase + clinopyroxene

The common feature is that all the possible accumulated crystal assemblages contain olivine – and this is consistent with the requirement that the magma must be accumulating Mg-rich minerals. Picrites are therefore basaltic rocks that contain a high proportion of olivine phenocrysts, plus or minus some cpx and/or plagioclase.

Part 2 (15 minutes): Observations of mineral textures

93-2 or R342A basalt

Examine the hand specimen. You should see two different phenocryst minerals in a glassy groundmass. What are the phenocryst minerals?

The phenocrysts are olivine and plagioclase. The texture of the hand specimen is glassy.

Examine the thin section. Before you put it under the microscope, hold the thin section up to the light and notice that it contains light and dark regions. You should also be able to see the phenocrysts.

Look at the light part of the thin section. In plane-polarised light the groundmass is light brown, but under crossed polars it is isotropic. It is therefore glass (which is an amorphous solid, and has isotropic optical properties). What does the presence of glass tell you about how quickly the magma cooled?

Basalt glass only forms when the cooling rate is extremely fast.

The glass contains phenocrysts of two different silicate minerals. What are they? Sketch the characteristic morphology (shape) of the phenocrysts. What does their size and shape tell you about the environment in which they grew? Comment on the likely ratio between the nucleation rate N and the growth rate G when the phenocrysts formed.

Olivines are typically euhedral. Plagioclases are typically tabular laths. The near-perfect crystal morphologies suggest that the minerals grew freely and unimpeded in the melt (liquid). The phenocrysts most probably grew in a magma chamber.

When the phenocrysts formed, G > N. This happens at small degrees of undercooling – that is, the magma was only a few degrees cooler than its liquidus temperature. The nucleation rate is low, so few crystals form. But the growth rate is moderate to high, so any crystals that do nucleate can grow to large sizes.

Some of the minerals in the light part of the thin section have a skeletal morphology, which is caused when the crystal edges grow faster than the faces. Sketch one of the skeletal olivine crystals.

Moving towards the centre of the thin section, notice the brown rims which appear around the phenocrysts. These rims are composed of radiating feathery crystals which nucleated on the phenocrysts during the rapid cooling of the basaltic melt. Comment on the likely ratio between the nucleation rate N and the growth rate G when the feathery crystals formed.

The feathery crystals form when G << N, at very high degrees of undercooling – that is, the magma temperature was dropped suddenly to several tens of degrees below the liquidus. The nucleation rate is very high, so lots of crystals can form. The growth rate is very low, so the crystals remain small.

Suggest a possible environment for the formation of this basalt.

The thickness of glass observed in these thin sections and hand specimens can only be produced during rapid cooling. There are no vesicles, which tells us that gas was not escaping from the magma. This means that it must have erupted under a confining pressure – most probably under water. (Not under air – otherwise there would be vesicles.) The most probable tectonic environment is a mid-ocean ridge. A subglacial eruption would also produce similar textures.

Part 3 (45 minutes): A full petrographic rock description **EXAM-STYLE QUESTION**

Thin section R93

The aim of this exercise is to produce a full petrographic description of thin section R93, including sketches. Use the guide to systematic description of igneous thin sections in the Practical Reference Handout p 3-4 to help you. This provides a suggested workflow for the observations you should make, and the level of detail you should aim for.

Before you start writing, spend a few minutes having a good look at the thin section. Think about the following questions, which are the basic starting point for any igneous rock description:

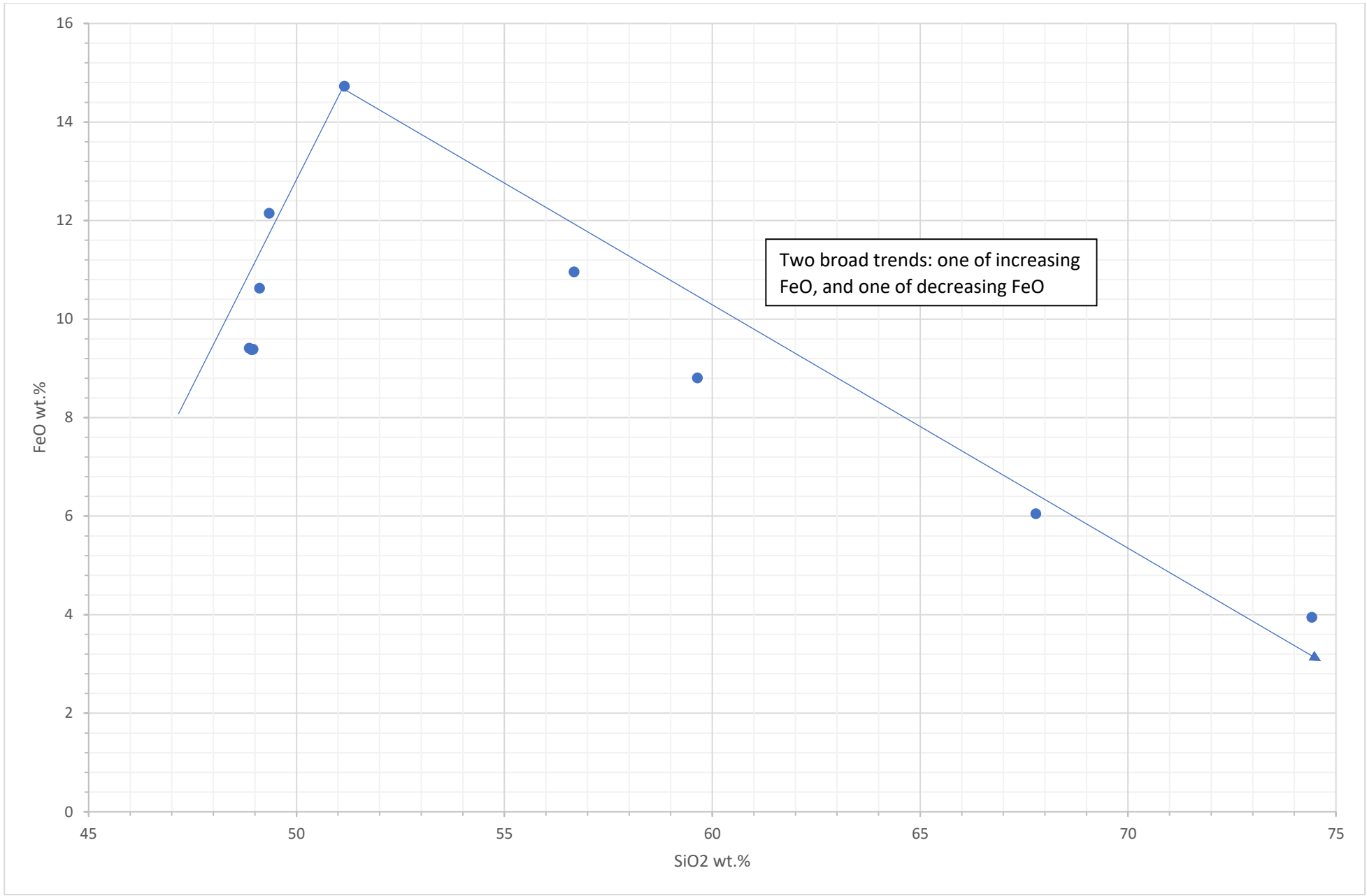
- What minerals can you identify?
- Which of these minerals are classed as major minerals? Which (if any) are accessory or secondary minerals?
- What is the grain size?

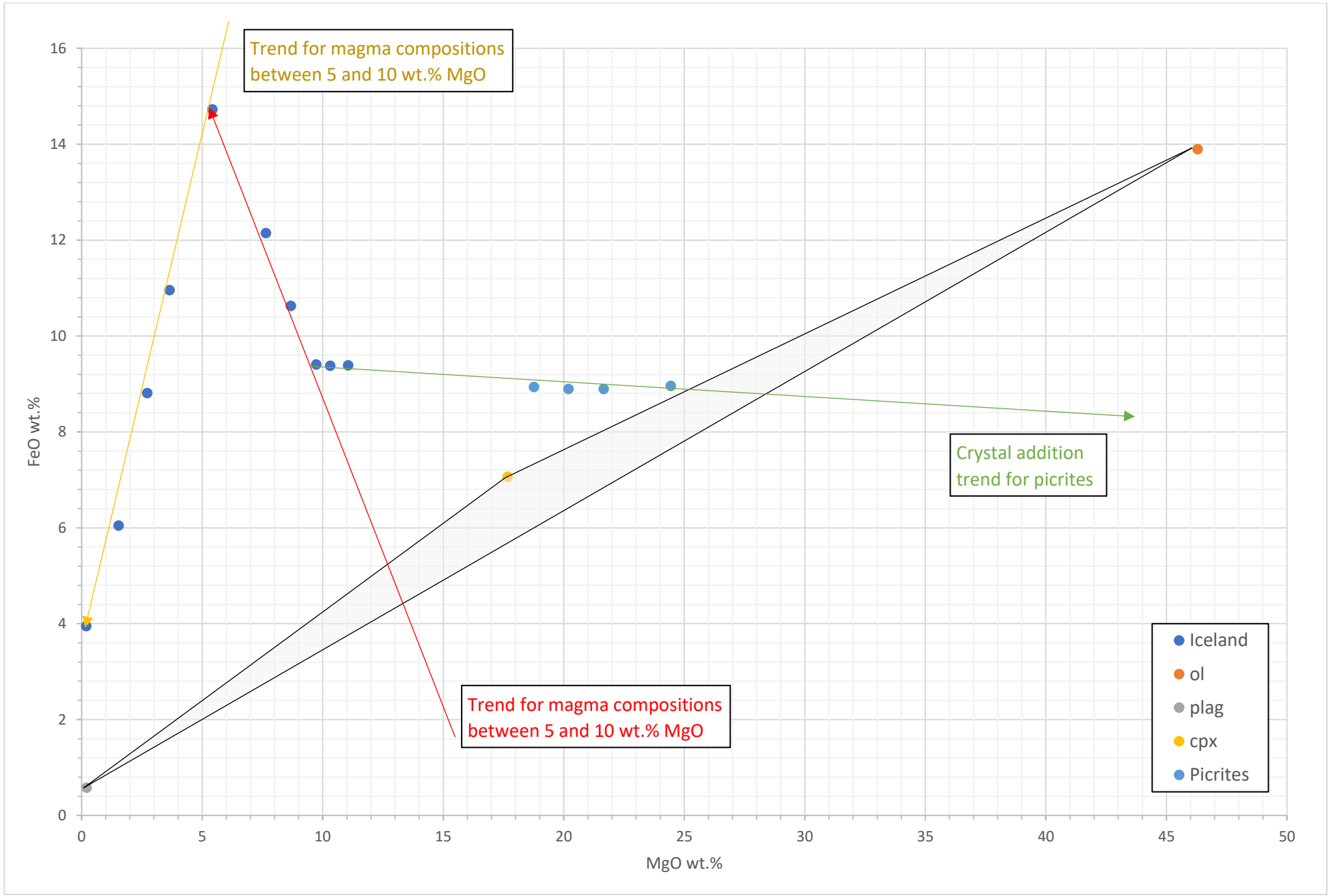
Work through each step systematically, making notes and sketches of your observations as you go. Skip over any steps where you don't yet know how to make some of the suggested observations, or if they are not relevant to this particular rock.

Add any interpretations to the end of your description and sketches. Concentrate on interpreting the cooling rate and the order of crystallization. (By the end of this unit you will be able to make more complex interpretations about the rock's history and tectonic context.)

Do not worry if you don't finish this exercise – you are practising the skills, and you will get faster as you become more experienced.

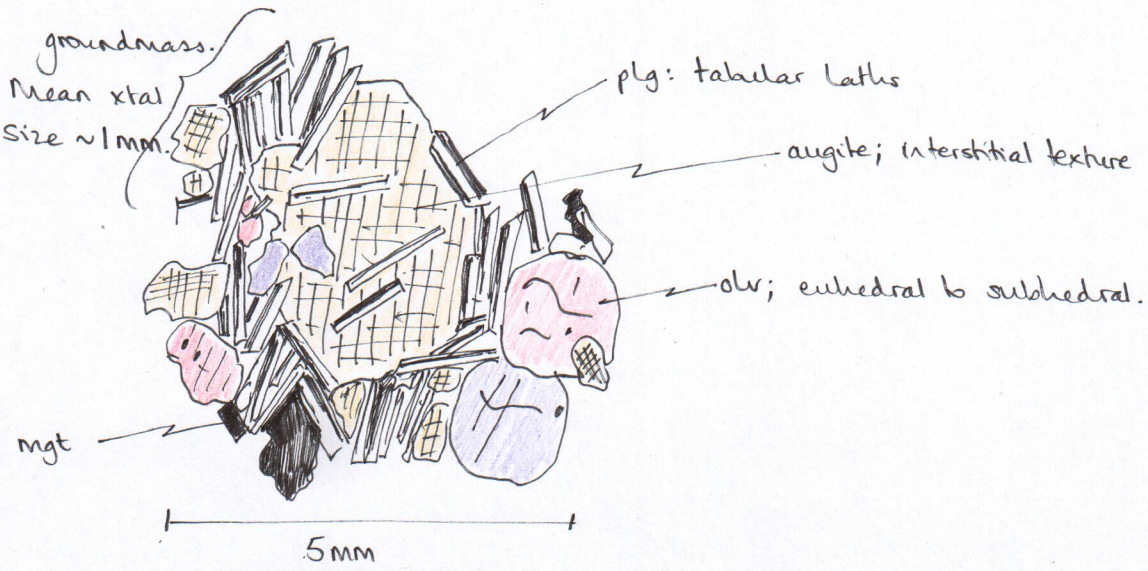
See separate file for a model petrographic description and interpretation


16
14
12
10
8
6
4
2
0
FeO wt.%
45
50
55
60
65
70
75
SiO2 wt.%
Two broad trends: one of increasing FeO, and one of decreasing FeO


Trend for magma compositions between 5 and 10 wt.% MgO
Crystal addition trend for picrites
Trend for magma compositions between 5 and 10 wt.% MgO
Iceland
ol
plag
cpx
Picrites
FeO wt.%
MgO wt.%
0
2
4
6
8
10
12
14
16
0
5
10
15
20
25
30
35
40
45
50

R93

Crystalline igneous rock. Typical grain size <1 – ~3 mm. No visible banding, preferred mineral alignment or internal structure.



Minerals

Olivine ~25%

- Typically 1–2mm (min ~0.2mm, max ~3mm)
- Subhedral, anhedral or euhedral ← Sometimes the 90° orthorhombic cleavages are visible



- High relief, curving cracks
- Occasionally has opaque inclusions. Cubic shape ⇒ spinel?
- Fresh appearance; no evidence of alteration

Augite (cpx) ~35%

- Range of xtal sizes, from <0.2mm to >3mm
- Largest xtals contain inclusions of plg (and occasionally olv)
- Interstitial texture
- In this sample cpx has high relief + faint dusty brown colour. Sometimes twinned.

Plagioclase ~35%

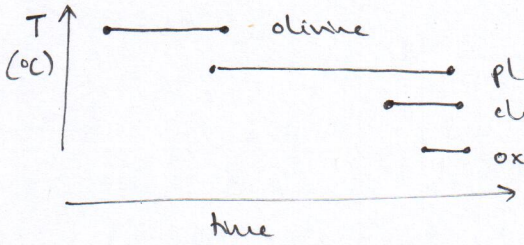
- Tabular laths, no more than 1mm in length. Main constituent of the groundmass.
- Found as inclusions within cpx but not within olivine
- Does not contain mineral inclusions.
- Twinned, unzoned.

Oxide (magnetite) ~5%

- Interstitial texture
- Grain size <0.2mm although occasionally larger crystals occupy interstices up to ~1mm in max. dimension

[Orthopyroxene (Not in every thin section)
- Similar appearance + texture to cpx in PPL
- Straight extinction, low 1st order birefringence.]

Order of crystallization.



- Olivine first: euhedral to anhedral shapes; no inclusions of the other minerals
- Plagioclase laths start crystallising next
- Pyroxenes crystallize in interstitial spaces between plagioclases. Plg + px crystallize together to form groundmass.
- Oxides crystallize in interstitial spaces.

Texture

OPHITIC: plag chadacrysts are enclosed by cpx oikocrysts

Pristine rock; no visual evidence of secondary minerals formed through an alteration process. [So it has not spent a long time in a wet environment.]

Rock name:

OLIVINE DOLERITE (Olivine gabbro also acceptable).

↑ type mineral

↑ root name implied by medium (to coarse) grain size + essential minerals plag + cpx

Interpretation.

An intrusive rock. Grain size suggests an intermediate cooling rate, so this could be a sample from a dyke or a thin sill. (ie. NOT a large magma chamber). Or it could be from the interior part of a thick lava flow.
Mineralogy is consistent with an olivine tholeiite magma, and a likely tectonic setting is at a mid-ocean ridge. Alternative – ocean island → this would be consistent with the lack of 2° alteration.
It's a relatively primitive magma because it still contains a high modal proportion of olivine.

[The rock is from Hawaii]